Technext

TECHNICAL REPORT 02

EnergieBee core Timeline & forecast

Data states, methods & provenance

Prepared by Technext
Source document · 16 October 2026
Report edition · 19 October 2026

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Original technical content and measurement dates are preserved. Diagrams appear on dedicated sheets sized for their detail. The editable Mermaid definitions remain in the source Markdown.

REPORT OVERVIEW

Prototype `energiebee-core-prototype` · Build method `v3-stock-follows-charge` · Forecast model v3 · Prepared by Technext · 16 October 2026

This document explains:

1. **How the half-hour timeline is created** from the providers' data.
2. **What `final`, `provisional`, `missing` and `forecast` mean.**
3. **How the 48-hour forecast works.**
4. **Where every number comes from, and what it affects** in the timeline, the daily table, the forecast, the tariff comparison and the pages.
5. **How the values the core learns from data are derived** (detection methods).
6. **What is still assumed, what are reporting choices, and what was removed.**

The rule the core follows: no number is shown unless it comes from a provider's data, is calculated from that data by a method described here, or is cited from a named public source. When none of these applies, the value stays empty (`null`), the half-hour stays unpriced, or the step is skipped with a reason. **Nothing falls back to a default.**

Examples use the real home **10 Acresbrook**: Sunsynk plant 292543, panels 6.5 kWp (Sunsynk), tilt about 20° facing about 60° east-north-east (estimated from the home's output, section 7), battery 11.0 kWh (measured, section 5.5); Octopus region G, Intelligent Octopus Go, gas zone NW.

1. How the timeline is created

1.1 The idea

Every provider measures something different, on its own clock, in its own units and with its own sign conventions. No page ever calls a provider. Each provider has an **adapter** that applies that provider's **contract**: its time zone, signs, units, how it shows missing data, and how it revises data. The adapter:

- stores the untouched payload (**raw**);
- writes **facts**, which are append-only and versioned: a changed value is a new version, and the old one is marked superseded.

A deterministic **build engine** turns facts into one row per UTC half-hour (`site_hh`) and one row per local day (`site_day`). The pages read only these rows, plus the forecast. The same facts and the same method always give the same numbers.

ENERGIEBEE CORE / FIGURE 01

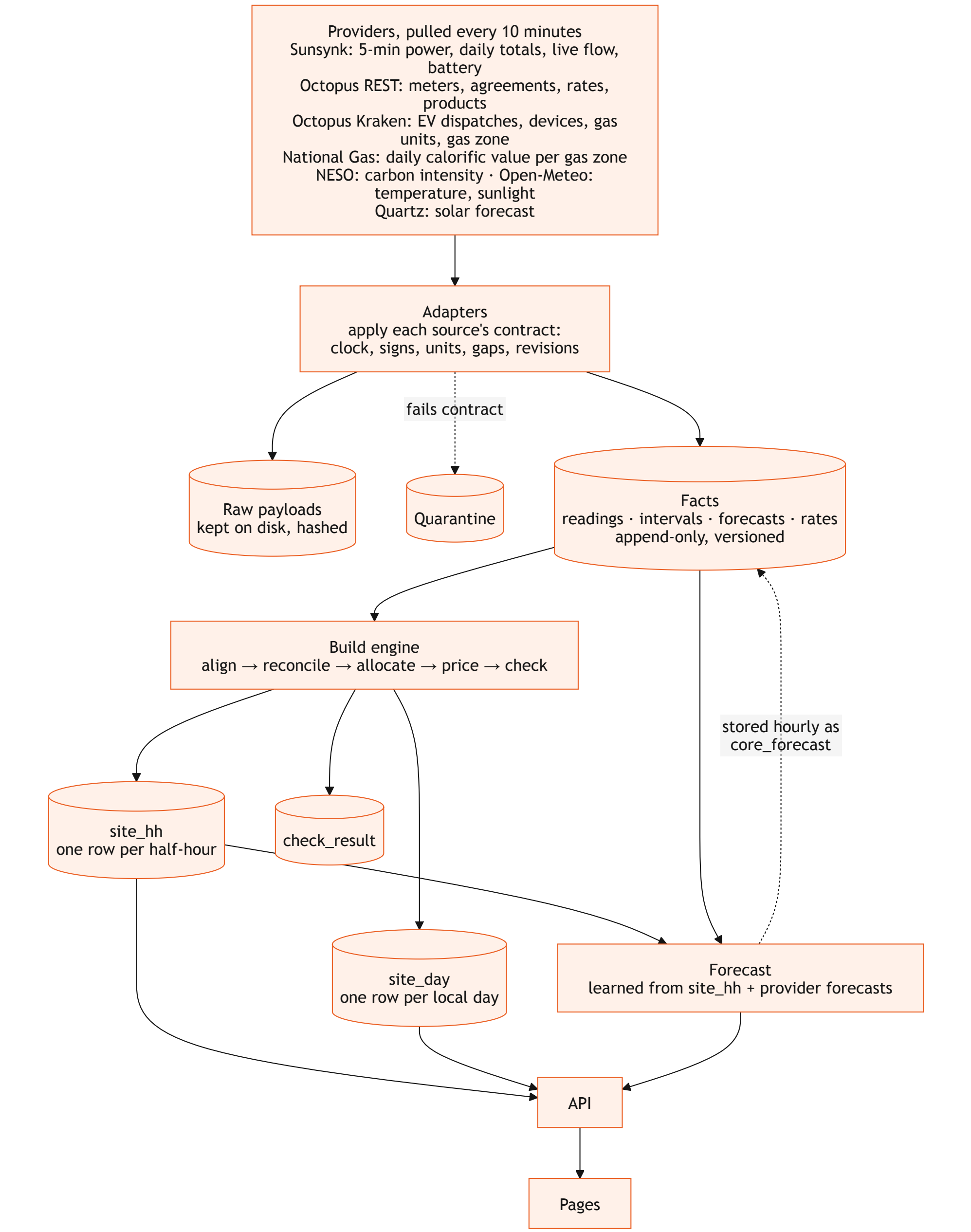


Figure 01. From providers to pages

1.2 The sync job (every 10 minutes)

Each step succeeds or fails on its own; a failure is shown on its connection.

1. **Sunsynk:** the named plant (`SUNSYNK_PLANT_ID`; this login sees 17 plants).
   - 5-minute series for yesterday and today (14 days on first run);
   - the month's daily totals and the live flow;
   - plant details: time zone, kWp (used only when `.env` has none), location;
   - the battery reading (capacity in Ah, voltage, lifetime counters), and a measured capacity check (section 5.5).
2. **Location:** postcode → coordinates (postcodes.io), Octopus grid supply point, NESO region. With no postcode, the nearest postcode to the inverter's registered location is used. Here both give BB12 9BY.
3. **Weather and carbon:** Open-Meteo temperature, cloud and horizontal sunlight, refilled to 15 days until 14 days are held; NESO regional carbon.
4. **Octopus:**
   - the account (meters, profile class, annual estimates, tariff agreements);
   - gas units and gas zone (LDZ) from Kraken;
   - meter readings with a 4-day look-back (14 days the first time);
   - rates for the tariffs in force, plus, when no time-stamped IOG prices are held, those of the home's earlier IOG tariffs (section 5.4);
   - Kraken dispatches and devices.
5. **Gas calorific value:** National Gas's daily value for the meter's gas zone (section 5.3).
6. **Comparison tariffs:** Agile and Agile Outgoing for the region (public prices).
7. **Quartz:** only when panel size, tilt and direction are known. It runs with the estimated angles (section 7). Recent output is not sent, because that made Quartz overshoot (section 3).
8. **Build:** rebuilds from the earliest half-hour whose facts changed in this sync (and always today), continuing from the battery stock saved at the end of the previous day. Never before the home's first inverter or meter data.
9. **Forecast record:** the forecast's house use, grid import and solar per half-hour are stored once an hour as `core_forecast`.

Fetching again is harmless:

- an identical payload hash is skipped;
- an identical fact is skipped;
- a changed fact becomes a new version.

1.3 The build engine (method `v3-stock-follows-charge`)

The build engine runs over every UTC half-hour of every local day. A local day has 46, 48 or 50 half-hours, depending on the clock change.

Step 1: Align (power → energy)

- Each 5-minute power sample is held until the next one (at most 5 minutes) and converted to kWh, split exactly at half-hour boundaries.
- **Clock changes.** Sunsynk gives each sample as a local "HH:mm" for the plant's day. Samples are taken in order and each gets the first matching instant after the previous sample. When clocks go back (last Sunday of October), 01:00–01:59 appears twice: the first run is BST, the repeat GMT, and both are kept (a 50-half-hour day). When clocks go forward, 01:00–01:59 does not exist and any sample with such a time is dropped (a 46-half-hour day). Tested with simulated payloads for 25 Oct 2026 (300 samples, all kept, in order) and 28 Mar 2027 (12 skipped).
- Each closed day is then scaled to the inverter's own daily counter for that flow.
- The factor is limited to 0.6–1.6 so that a corrupt counter is ignored. At 10 Acresbrook the factors run 0.94–1.20 (medians 1.01–1.04).

Step 2: Reconcile (one value per flow)

Value	Rule
Grid import / export	Smart meter where it has a reading, otherwise the inverter clamp (the half-hour is then provisional)
House use	The inverter's measured load, limited to what the sources supplied
Balance	solar + import + battery out − export − battery in
Inverter loss	balance − measured load (about **3.0 kWh/day** here)
Gas	Gas meter reading as delivered (m³ here), converted to kWh with that gas day's published calorific value (section 5.3); unknown until it is published
EV	Kraken completed dispatches

Step 3: Allocate (who fed whom)

Flow	Rule
solar → home	min(solar, home)
solar → battery	min(solar left, battery in)
solar → grid	min(solar left, export)
battery → home	min(battery out, home not yet covered)
battery → grid	min(battery out left, export left)
grid → home	min(home still not covered, import)
grid → battery	min(battery in not covered by solar, import left)
each source → loss	what each source supplied beyond the flows above

Step 4: Price

- **Tariff:** the agreement in force for that meter at that half-hour (from the Octopus account).
- **Rate band:**
  - Intelligent Octopus Go: night inside the off-peak window **learned from Octopus's time-stamped IOG prices** (section 5.4), or in a completed smart-charge dispatch half-hour that is within the car's daily allowance (first 12 half-hours, i.e. 6 hours, per midday-to-midday period, as Octopus states; section 5.4); day otherwise.
  - Economy 7: **not priced**, because no source gives the switch times.
  - Agile and flat tariffs: the single published price.
- **Payment method:** a price is used only if it applies to any payment method, or every payment method has the same price. The API does not say how the customer pays.
- **VAT:** from the dated table (section 4.3).
- **Standing charge:** the daily charge divided by the number of half-hours in that local day.
- **Battery stock, first in first out:**
  - the opening layer is the energy at the first charge-level reading, with **cost unknown** (section 5.2);
  - grid charging adds a layer at price ÷ **measured** round-trip efficiency (section 5.1);
  - solar charging adds a layer at 0p (no fuel cost);
  - energy drawn from the opening layer has no stored cost (`null`);
  - **the stock follows the measured charge level.** At the end of each half-hour the layers are scaled together so they add up to charge level × capacity. Energy in × efficiency − energy out drifts from what the battery holds, because efficiency and capacity are measured, not exact. Before this step (method `v2-measured-load`) the modelled stock ran dry on 9, 12 and 15 Oct while the battery still held 20%. Those half-hours had no stored cost, so the 7-day blended cost showed as unknown. Scaling keeps each layer's share, so the price mix is unchanged. Bill and saving are not affected.
- **Bill** = every imported kWh × price (house + battery charging + inverter loss) + standing charge + gas − export credit.
- **Grid-only** = house use × price + standing + gas.
- **Saving** = grid-only − bill.
- **Carbon** = import × regional intensity; avoided = house use × intensity − that.

Step 5: Check and set the state

Check (daily)	Pass when	What it means
Completeness	no half-hour missing	Every half-hour has a grid reading
Rate coverage	every half-hour priced	No unpriced energy
Energy balance	the inverter's use (the `*_to_loss` flows) within ±max(1 kWh, 50%) of this home's median over the 30 days before; judged from the 8th day	The house, exports and charging account for what came in. A fixed share of house use was dropped: the inverter's standby use is roughly constant, so it warned on 11 of 15 quiet days.
Meter vs clamp, import and export	daily total within 5% (or 0.2 kWh)	Two devices measuring the same connection agree; the clamp is a sound fallback

Check (daily)	Pass when	What it means
Counter fit: solar, house, import, export, battery in, battery out	within 10% (or 0.2 kWh) of the inverter's own daily counter	The 5-minute samples are complete. The counters read in 0.1 kWh steps, hence the 0.2 kWh floor.
Gas calorific value	every gas half-hour converted	National Gas has published the value for that gas day
Battery efficiency	measured	Shows the value and how it was measured
Battery stock	—	Warns that opening energy has unknown cost

Checks run on completed days only; today is still filling in.

Rebuild: the battery stock carries from one half-hour to the next. Each day saves its closing stock (`site_day.battery_stock`), so a rebuild can start at any day and continue from the day before. A normal sync rebuilds only from the earliest changed day (1 day, 13–18 ms). A full rebuild (Data & trust → Rebuild, about 0.2 s for 15 days) gives byte-identical rows.

1.4 Worked example: Fri 9 Oct, 11:00–11:30 (10 Acresbrook)

Facts in:

- Sunsynk 5-minute samples: solar 633 → 1,316 W; load 1.0–5.7 kW; battery 0.5–4.1 kW out; charge level 54% → 42%.
- Octopus meter: import 0.215 kWh, export 0.009 kWh.

Result	Value	Comes from
Solar	0.560 kWh	Sunsynk 5-min × that day's counter fit 1.126
Battery out	1.552 kWh	Sunsynk 5-min × fit
Import	0.215 kWh	Octopus smart meter
House use	2.224 kWh	Sunsynk measured load
Inverter loss	0.094 kWh	balance − load
solar / battery / grid → home	0.560 / 1.552 / 0.112 kWh	allocation
Price	32.14p/kWh, 0% VAT	IOG day band (learned window), dated VAT
Battery energy's stored cost	12.4p	first in first out: overnight charge at 7.62p ÷ measured round-trip efficiency
Bill	**8.7p** = 3.6p grid→home + 3.3p grid→loss + 1.5p standing + 0.3p gas	
Grid-only	**73.3p** = 2.224 × 32.14p + standing + gas	
Saving	**64.6p**	
State	`final`; quality 129 = measured + revised	

2. What the data states mean

State	Meaning	What the user sees
final	Built from the smart meter (homes with a supplier account) with every check run. Changes only if a provider revises its data.	Normal figures
provisional	Built from the inverter's grid clamp because the meter reading hasn't arrived (Octopus is about a day late), or the day isn't over. Rebuilt automatically when the reading arrives.	"Estimate until your smart meter reports"
missing	No grid reading from any source.	A gap, never a zero
forecast	After now, from the forecast (section 3).	Faded bars after "Now"

ENERGIEBEE CORE / FIGURE 02

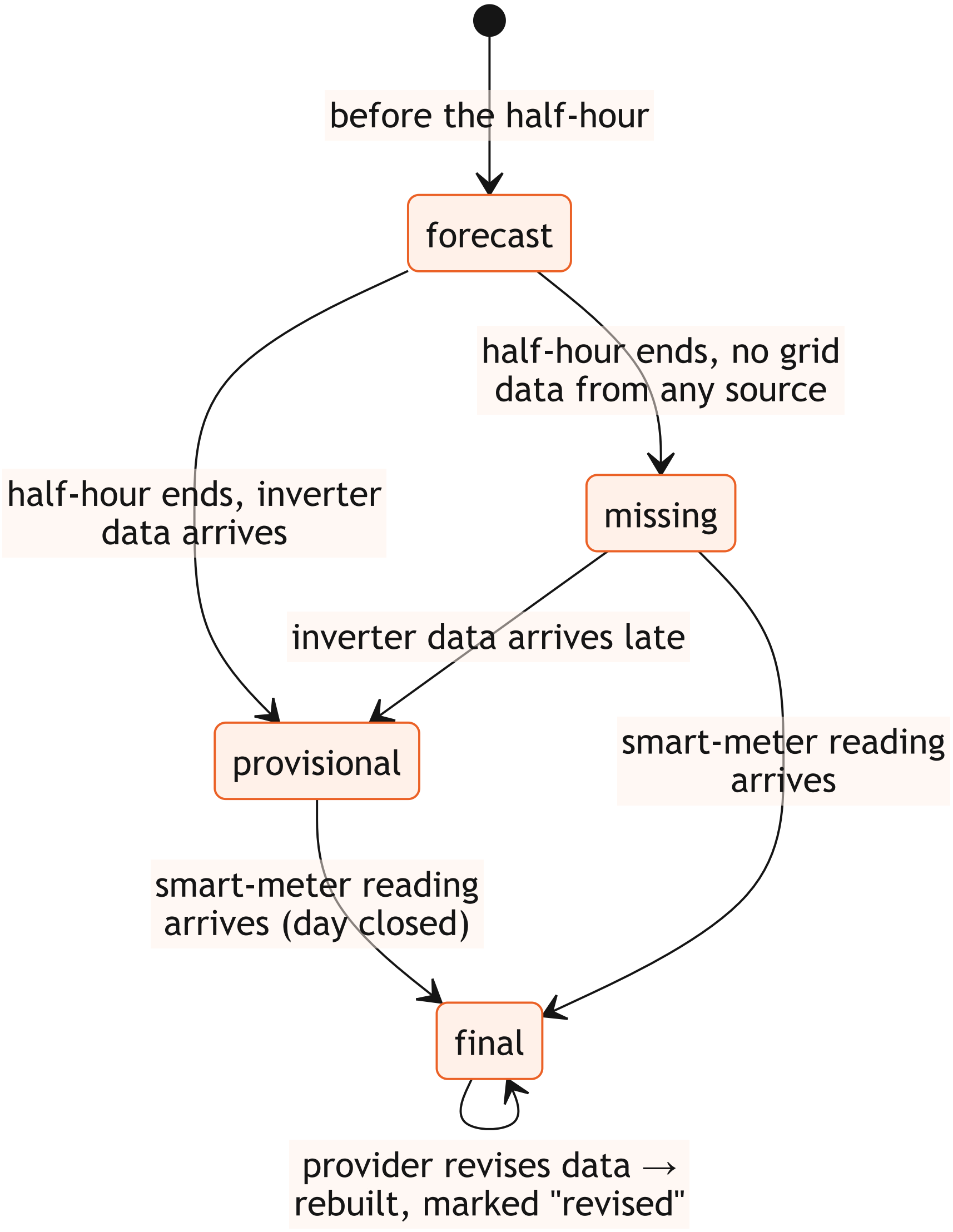


Figure 02. Data states and how a half-hour moves between them

Revised: once a final half-hour's numbers change, it is marked permanently (a dot on the Timeline). 566 half-hours here are revised. They were first built before the Octopus key worked, and corrected when the meter data, the account's tariff, gas units and measured values arrived.

Quality flags:

Flag	Meaning
measured (1)	grid from the meter
fallback (2)	grid from the clamp
estimated (8)	battery energy of unknown cost
partial (16)	inverter covered under 80% of the half-hour
suspect (32)	negative balance
check failed (64)	no price
revised (128)	changed after being final

Every half-hour's evidence page lists its lineage, every fact version and the raw payloads.

3. How the forecast works (model v3)

The forecast covers the next 48 half-hours × 2. Every parameter is learned from this home's last 14 days, or taken from a provider's forecast. Anything that can't be learned is left out, with the reason given in `model`.

Part	Method	Learned at 10 Acresbrook
Solar	Two methods; the one checked against this home is used. **Learned from this home:** Open-Meteo's forecast of horizontal sunlight × this home's own output per unit of sunlight for each hour of the day, fitted over the last 14 days. **Quartz** (Open Climate Fix), for the configured size and angles, is used only once 2 days and 2 kWh of its forecasts can be compared with actual output; it is then scaled by the observed ratio. The comparison uses Quartz's own forecast of record (the latest issue before each half-hour), never our stored forecast. If actual ÷ Quartz is outside 0.5–2, Quartz is not used: a gap that large means it was given the wrong panels, not that it needs scaling. On 16 Oct the panels made **5.78×** Quartz's forecast (6.5 vs 1.1 kWh, one day), with Sunsynk's 6.5 kWp and the estimated 20°/60° angles; the learned path stays in use. No half-hour is forecast above the most this home has produced in one (2.3 kWh).	Learned path, R^2 0.72 on 15 days. Quartz is checked from 18 Oct.
House use	Median of each half-hour of the day, from days of the same type (weekday or weekend), plus planned EV charging. A temperature adjustment is fitted (heating degree days, base searched 10–20 °C) and applied only if use rises in colder weather with $R^2 \geq 0.3$.	Weekday 11 days, weekend 4; temperature **not applied**: use does not rise in the cold
Battery	Starts at the latest charge level. Uses the observed charge range, half-hour limits and measured efficiency. Charges from the grid in the **learned charge window**, holds in the **learned hold window**, stores spare solar, and covers shortfalls otherwise.	20–100%; 1.94 kWh in / 1.55 kWh out per half-hour; efficiency 95.4%; charge 00:00–02:30 up to ~100%; hold 23:30–00:00 and 02:30–05:30
Prices	Published rates of the home's tariff agreements, with VAT. No price → no cost.	IOG 7.62p / 32.14p
Carbon	NESO regional forecast, latest issue.	Region 3

Part	Method	Learned at 10 Acresbrook
Results	Expected import = use − solar ± battery; export = spare solar. Cost = import × price − export × export price. Grid-only = use × price. Saving = the difference.	
Ranges	House use: 10th and 90th percentiles at each half-hour. Solar: spread of daily actual-to-fitted ratios (needs ≥ 3 days). Three battery runs: expected, low (low use, high solar) and high (high use, low solar).	Solar ×0.87–1.22 over 13 days (in-sample)
Accuracy	Our forecasts are stored hourly. Each finished half-hour of the last 7 days is compared with the latest forecast issued ≥ 12 h before it.	First scores once a whole day has been forecast 12 h ahead

Limits:

- The solar range is in-sample, so it does not include errors in the sunlight forecast itself.
- 24-hour range totals add up the half-hour bounds, so they are wider than a true daily spread.
- Agile prices beyond what is published are not estimated.
- The past dashed "forecast at the time" line on the Solar page fills in from 16 Oct onward. Earlier Quartz forecasts were deleted because they were made with invented panel angles (section 6.3).

3.1 Solar made against what the weather allowed

The Solar page sets each past half-hour's output against two yardsticks on one axis: the **forecast made in advance** (`site_hh.solar_forecast`, the latest issue before the half-hour started, dashed) and the output **expected for the weather that actually happened** (solid). The second is an expectation, not a measurement. It comes from `GET /api/sites/:id/solar-expected?period=` (`src/core/solar-expected.js`) and is computed on request, not stored.

Part	Method
Sunlight	Open-Meteo's horizontal sunlight for each past hour at the home's location (`interval_fact`, `ghi_wm2` / `open_meteo`, versions in force). These are Open-Meteo's model values for past hours, not a sensor at the home. The hour's mean W/m² gives kWh/m² for each of its half-hours, as in the forecast.
Response	The forecast's own fit (`learnSunlight`): for each local hour of day, Σ(solar × sunlight) ÷ Σ(sunlight²) over built half-hours (not `missing`) with sunlight above zero.
Window	The last 30 days, ending today. This is a reporting choice: it matches the Solar page's longest view. Because the window always ends today, a day gets the same expected value in the Today, 7-day and 30-day views.

Part	Method
Leave one day out	Day D is compared with a fit over the window's other days, never D. A day's own output cannot shape its yardstick, so a bad day shows up as bad.
Expected	Sunlight × the ratio for that hour. Zero sunlight gives 0. There is no value (null, never 0) when a half-hour has no sunlight fact, when no other day had sun at that hour, or when fewer than 3 other days have both solar and sunlight.
Made vs expected	Actual ÷ expected, over the half-hours that have both. A gap on either side is therefore never counted as zero on one side only. It is given per day, for the period, and as a low–high range over whole days.
Check	Every daylight half-hour of the window is compared with the fit made without its own day, giving an out-of-sample R^2 and mean absolute error.

At 10 Acresbrook (copy of 16 Oct, early afternoon): 15 days in the window. The out-of-sample check gives R^2 0.69 and a mean error of 0.22 kWh over 318 daylight half-hours. Over the last 7 days the panels made 78.9 kWh against 73.5 kWh expected (107%). Whole days ranged from 78% to 135%.

Limits:

- Open-Meteo's past sunlight is modelled for the location. Cloud over the home that the model missed shows up as over- or under-performance.
- The fit is linear per hour of day. It reflects the panels' angles and shading as they were over the window. It does not know the inverter's limit, so on the brightest half-hours the expected value can be more than the inverter could deliver.
- Days before Open-Meteo's history starts have no expected value. They show as gaps, not zeros.

4. Where every number comes from, and what it affects

4.1 Provider data

Provider · data	Used for	Lands in	Affects
Sunsynk 5-min (PV, load, grid clamp, battery, charge level)	Half-hour solar, house use, battery in/out, charge level; grid fallback	`reading` → `site_hh.solar, home, battery_*, soc_end, grid_*` when provisional	Every page; flows; inverter loss; battery valuation; forecast learning
Sunsynk daily totals	Counter fit per day	lineage `fit`	All Sunsynk-based energy (±0–20%); counter-fit check
Sunsynk live flow	"Right now"	`reading` (live)	Home flow diagram; forecast start charge level
Sunsynk plant	Time zone, location; kWp only if `.env` has none (it reports 6.5 kWp; `.env` says 5.0)	`site`	Day boundaries; postcode (via location)
Sunsynk battery reading	200 Ah × 51.5 V = 10.3 kWh; lifetime counters 8,173.1 kWh in / 7,873.7 kWh out	`connection.meta`, raw #533	Cross-check of capacity and efficiency (section 5.5)
Octopus meters (import, export, gas)	Billed energy	`interval_fact` → `site_hh.grid_*`, `gas`	Bill, saving, state (final vs provisional), meter-vs-clamp check
Octopus account	Meters, MPAN/MPRN, **profile class**, annual estimates, **tariff agreements**	`device`, `tariff_agreement`	Which price applies when; typical-home comparison (profile class)
Octopus rates	Unit and standing prices, VAT-exclusive	`rate_interval`	Every cost; IOG window learning; tariff comparison
Octopus products	Current Agile and Agile Outgoing codes	`connection.meta` (compare)	Tariff comparison only
Octopus Kraken	EV dispatches and devices; **gas `consumptionUnits` and gas zone (`ldz`)**	`interval_fact` (EV), `device`	EV energy, IOG dispatch pricing, which gas readings are m³, which calorific value applies
National Gas Data Portal	Daily calorific value for the gas zone (item "Calorific Value, LDZ(NW)", PUBOB4512)	`interval_fact` (one row per gas day)	Gas kWh, gas cost, bill
NESO	Regional carbon intensity (history = NESO's forecast)	`interval_fact`, `forecast_fact`	Carbon emitted/avoided, greenest window

Provider · data	Used for	Lands in	Affects
Open-Meteo	Temperature; horizontal sunlight; sunlight on the panels (only with tilt and direction)	`reading`, `interval_fact`, `forecast_fact`	Timeline weather row; temperature fit; sunlight-based solar forecast; Solar page "expected for the weather" (section 3.1)
Quartz	Solar forecast for the panels (with size, tilt and direction)	`forecast_fact`	Solar forecast; Solar page forecast line
`.env`	Postcode; panel tilt 20° and direction 60° **estimated from the home's output** (section 7); panel size and battery left empty so the providers' values are used	`site`	Quartz request, sunlight on the panels
postcodes.io / Octopus GSP / NESO region	Postcode, region keys	`site`	Which prices and carbon series apply
Ofgem TDCV 2026 (public document)	Typical-home figures: 2,500 kWh electricity (profile class 1), 3,400 (class 2), 9,500 kWh gas	page constant, cited	"How you compare" on My energy only

4.2 The core data matrix: each serving column and its inputs

`site_hh` columns	Inputs	Methods (section)
`solar`, `battery_charge`, `battery_discharge`	Sunsynk 5-min, daily totals	align + fit (1.3)
`grid_import`, `grid_export`	Octopus meter; else Sunsynk clamp	reconcile (1.3)
`home`	Sunsynk load, limited to the balance	reconcile (1.3)
`loss`, `*_to_loss`, `loss_cost`	balance − load; allocation; price	1.3
`gas`, `gas_cost`	Octopus gas meter (m³) × 1.02264 × National Gas calorific value ÷ 3.6; gas tariff	5.3
`ev`	Kraken dispatches	1.3
`soc_end`	Sunsynk charge level	—
seven flows `*_to_*`	all of the above	allocate (1.3)
`import_rate_ex`, `import_vat`, `export_rate`	Octopus agreements + rates; learned IOG window; VAT table	1.3, 5.4, 4.3
`grid_to_home_cost`, `grid_to_battery_cost`, `export_credit`, `standing_cost`, `total_cost`, `grid_only_cost`, `saving`	flows × prices	1.3

site_hh columns	Inputs	Methods (section)
battery_to_home_cost	first-in-first-out stock; measured efficiency; opening layer	5.1, 5.2
carbon_*	NESO	—
outdoor_temp	Open-Meteo	—
solar_forecast	latest forecast issued before the half-hour (Quartz, else core_forecast)	3
state, quality, lineage	all checks	1.3, 2

site_day is the sum or extreme of these per local day, plus self-consumption, self-sufficiency, blended cost (unknown when stored cost is unknown), peak import, base load, charge range, cycles and integrity.

4.3 VAT

Fuel · direction	Rate	Source
Electricity import from 30 Sep 2026 23:00 UTC	0%	Octopus prices: VAT-inclusive equals VAT-exclusive
Electricity import before that	5%	UK VAT law (outside the current data window, so no effect on today's figures)
Electricity export	0%	Octopus Outgoing prices
Gas	5%	Octopus gas prices: inclusive = exclusive × 1.05

5. How the learned values are derived (detection methods)

5.1 Battery round-trip efficiency

- **Formula, per day:** energy out ÷ (energy in − change in stored energy) over the 30 days ending that day. Energy in and out come from the inverter's own daily counters; stored-energy change = Δ charge level × capacity. A day's value depends only on facts up to that day, so it is the same whether the day is rebuilt alone or in a full rebuild. Stored per day in `site_day.battery_eff`.
- **Result on 16 Oct:** 118.4 kWh out ÷ (129.1 kWh in − 3.2 kWh change in stored energy) = **94.0%**. The forecast's own measure, from 5-minute data over unbroken half-hour pairs, is 95.4%. The two methods differ by 1.4 points; the inverter's counters are used for valuation.
- **Affects:** the stored cost of grid-charged battery energy (price ÷ efficiency), and therefore `battery_to_home_cost`, blended cost, stored-energy p/kWh on the Battery and Breakdown pages, and forecast battery losses.
- **Without enough data:** no loss is applied and the check says "not measured".

5.2 Opening battery stock

- The energy in the battery at the first charge-level reading (70% × 11.0 kWh = 7.7 kWh on 2 Oct 00:00) becomes the first stock layer, **with unknown cost**.
- **Affects:** `battery_to_home_cost` is `null` for half-hours drawing from it (2–3 Oct only; 34 half-hours, 6.4 kWh), and blended cost is unknown for periods that include them. The bill and the saving are **not** affected: battery charging is paid when it happens.

5.3 Gas units and calorific value

Unit. Decided once per meter and stored with its source:

1. **Octopus states it:** Kraken `consumptionUnits`. This meter reports **m³** (SMETS2).
2. Only if Octopus says nothing: a half-hour above 2.5 can only be kWh; otherwise the yearly rate is compared with the meter's annual estimate (19,270 kWh). Without data the unit stays unknown and the readings are not used.

Readings are stored **exactly as delivered** (`gas_m3`), never pre-converted.

Calorific value. GB gas is billed on volume × the calorific value (CV) of the gas delivered in that zone on that day:

- the meter's gas zone comes from Octopus (Kraken `ldz` = **NW**, exit zone NW1);
- National Gas publishes the daily CV per zone on its Data Portal. The data item is found by its name, "Calorific Value, LDZ(NW)", which is id **PUBOB4512**;
- each value applies to one **gas day, 05:00 to 05:00 GB local time** (UNC Modification 0461; SI 2015/953);
- values at 10 Acresbrook, 1–15 Oct: **38.9–40.6 MJ/m³**.

Conversion per half-hour: kWh = m³ × **1.02264** (volume correction, Gas (Calculation of Thermal Energy) Regulations 1996) × CV for that gas day ÷ 3.6.

Gas day	m³	CV (MJ/m³)	kWh
2 Oct	1.114	40.44	12.79
11 Oct	5.761	39.40	64.38
14 Oct	3.360	39.06	37.31

- **Until a day's CV is published** (the next morning, about 10:30), that gas day's kWh stay empty. The daily check `gas_calorific_value` says how many half-hours are waiting. Nothing is converted with a typical value.
- **Affects:** `gas`, `gas_cost`, the bill, grid-only cost, gas on My energy and Breakdown.
- **History:** earlier versions converted with a fixed 39.5 MJ/m³ and are kept as superseded facts. Against them, the 7-day gas total moved from 193.5 to 191.7 kWh (−0.9%), with individual days moving by up to about 2.5%.

5.4 Intelligent Octopus Go off-peak window

- **Problem:** the current tariff (IOG-SMB-FIX) publishes day and night prices without times.
- **Method:** Octopus's time-stamped IOG price lists in the same region mark the window as the cheaper rows. Here, the 17 off-peak rows (of 33) of this home's previous tariff E-1R-INTELLI-FIX-12M-25-07-30-G give **23:30–05:30 local**.
- **Confirmation:** the meter data shows the battery charging and the house drawing from the grid in exactly that window.
- **Also stated by Octopus:** "a super low price between 11:30pm - 5:30am for your whole home" (Octopus help article *How does Intelligent Octopus Go work*).
- **Smart-charge slots:** "Your car gets up to 6 hours of off-peak charging per day (measured midday to midday). If we schedule your car to charge outside of the 23:30–05:30 window, your home also gets the off-peak rate for that half-hour when it's within your car allowance." (Octopus blog, 7 May 2026). The core applies exactly this: the first 12 dispatched half-hours in each midday-to-midday period are priced at the night rate for the whole home. There are no dispatches here today, so it has no effect yet.
- **Affects:** the register (day/night) for every half-hour, and so every import cost, the tariff comparison and the price strips.
- **Without such rows:** IOG half-hours stay unpriced and the rate-coverage check fails.

5.5 Battery capacity

The core uses capacity mainly to turn the battery's charge level (%) into kWh, so the figure it uses is **measured kWh per 1% of charge**.

Figure	Value	Source	Used?
Installer's rated capacity	not known	`BATTERY_KWH` in `.env` (empty)	would take priority
Measured from use	**11.0 kWh per 100%** = 104.6 kWh discharged over 951 points of charge drop, 1,724 discharge-only 5-minute steps in 30 days	Sunsynk 5-minute power and charge level (`src/core/derive.js`)	**yes**

Figure	Value	Source	Used?
Battery's own reading	200 Ah (`correctCap`) × voltage at that moment: 10.3, 10.6, 10.7 kWh on three readings	Sunsynk battery payload (raw kept)	cross-check only; it drifts with voltage

- **Lifetime counters** (Sunsynk): 8,173.1 kWh charged, 7,873.7 kWh discharged, 96.3% over the battery's life, against 94.0% over the last 30 days (5.1).
- **Affects:** kWh of stored energy, cycles, the stored-energy term in efficiency, opening stock, and forecast battery limits.

5.6 Location and region

- Postcode **BB12 9BY**, from the nearest postcode to the inverter's registered coordinates. It is confirmed by the Octopus supply address and set in `.env`.
- That gives Octopus region G (prices) and NESO region 3 (carbon).

5.7 Forecast detections

Detection	Rule
Grid-charge window	Half-hours where the battery took > 0.05 kWh from the grid on ≥ 50% of the last 14 days. Target = median charge level at the window's end.
Hold window	Half-hours where the house drew from the grid while the battery had charge above its observed minimum and did not discharge, on ≥ 50% of days.
Solar response to sunlight	For each local hour of the day: solar kWh per unit of horizontal sunlight over 14 days (least squares through the origin).
Day type and temperature	Section 3.

5.8 Typical-home figure

- **Rule:** the import meter's **profile class from Octopus** (1 = single rate, here) selects Ofgem's 2026 medium figure: 2,500 kWh. Gas uses 9,500 kWh.
- **Source:** Ofgem, *Review of typical domestic consumption values: decision* (May 2026), Table 1, in force from 1 July 2026.

6. What is still assumed, reporting choices, and what was removed

6.1 Still assumed

None of the values that feed costs or energy is assumed any more. The three that were assumed in the previous version now have sources:

Was assumed	Now
Gas calorific value 39.5 MJ/m³	National Gas daily value for the meter's zone, from Octopus's `ldz` (5.3)
IOG: whole home at night price during a smart-charge dispatch	Octopus's published rule, including the 6-hour midday-to-midday allowance (5.4)
Battery capacity from one voltage reading	Measured kWh per 1% of charge (5.5)
Panel tilt and direction (placeholder 30°, south)	Estimated from the home's output (section 7)
Panel size (placeholder 5.0 kWp)	Sunsynk plant record, 6.5 kWp. Consistent with output peaking at 5,345 W, above 5 kW in 20 samples on 5 days.

What is estimated rather than measured directly: the panel tilt and direction (section 7). Installer values from the MCS certificate replace them when available.

6.2 Reporting choices (methods, not data)

These shape how results are summarised, never the measured values:

- check tolerances (10%, 5%, 10%);
- counter-fit limits 0.6–1.6;
- the 14-day learning window;
- 10th/90th percentiles;
- "≥ 50% of days" for learned windows;
- the 0.05 kWh activity threshold;
- the 2-hour cheapest/greenest window;
- price bands (0–10p, …);
- the comparison's 16:00–19:00 window;
- "last 7 days" projections on My energy;
- the 12-hour lead for accuracy;
- the 0.5–2 range for using Quartz (section 3);
- the check tolerances in section 1.3 (5%, 10%, the 0.2 kWh floor, the energy-balance band and its 7-day start);
- when each source counts as late (section 8.2);
- the 30-day window for solar expected for the weather (section 3.1).

Each is named where it is shown.

6.3 Removed (invented without a source)

Removed	Where it was	Replaced by
Appliance list and kWh per run (dishwasher, washing machine, tumble dryer, 10 kWh car top-up)	My energy	Price difference per kWh only
"Miles added" at 3.5 miles/kWh	EV page	—
Panel tilt 35°, facing south (put in `.env` by me)	Quartz forecast, sunlight on panels	Forecasts made with them deleted; the owner has since given the real values (30°, 180°)
Demo home (fully simulated house)	Whole app	Removed from code, database and picker
Advice text not backed by data (thermostat 10%, "fridge, router", "normal for an inverter", appliance timing, Agile behaviour claims, "reconnect the car")	Insights, comparison, My energy	Statements of measured or provider-reported facts only
Ofgem 2023 typical figures from memory	My energy	Cited 2026 figures by profile class
Battery efficiency 0.95	Build, forecast	Measured (5.1)
Gas calorific value 39.5 MJ/m³	Gas conversion	National Gas daily value (5.3)
Opening battery energy at 0p	Build	Unknown-cost layer (5.2)
IOG window typed into code	Pricing	Learned from Octopus rates (5.4)
Economy 7 hours	Pricing	Not priced
"Pays by direct debit"	Pricing, comparison	Payment-independent prices only; Flexible removed from the comparison
Quartz 4 kWp fallback	Solar forecast	No call without real values
Forecast battery 3.6 kW cap, 10%/98% limits, ±25% solar range, 15.5 °C base	Forecast	Observed limits; no range without data; searched base
"About 10% round-trip loss"	Battery page	Measured loss shown

7. Panel tilt and direction

The installer's values are not known, and Sunsynk does not report angles. They are **estimated from the home's own output** (`src/core/derive.js`, `estimatePanelAngles`) and written to `.env` with a note saying so.

Method:

1. Hourly actual solar for the last 14 days, from `site_hh`.
2. For each candidate tilt and direction, Open-Meteo's sunlight on a panel at that angle for the same hours. The coarse grid is tilt 10–60° × direction 90–270°, refined around the best (71 candidates).
3. For each candidate, fit actual = k × sunlight through the origin over sunny hours (sunlight ≥ 50 W/m²; 99 hours) and score it by R².

Result:

Tilt · direction	R²
20° · 60° (east-north-east), best	**0.602**
15–25° · 60–90°	0.57–0.60
30° · 90° (east)	0.510
30° · 180° (south, the earlier placeholder)	0.400
30° · 270° (west)	−0.146

- **The direction is clear; the exact degree is not.** East to east-north-east at a shallow tilt fits; south and west do not.
- **One direction is right for this home.** Sunsynk's live readings show both strings producing almost the same power through the morning (for example 1,959 W and 1,993 W at 10:12), so both face the same way.
- **Limits:** 14 October days of mixed cloud; afternoon shading would look like an easterly direction.

Replacing the estimates with the installer's figures. The UI shows each detail, its value, where it came from and whether it is confirmed: **Data & trust** → **Your system's details**. The Solar page also shows a notice while the angles are estimates. The steps:

1. **Get the installer's handover pack.** It contains the MCS certificate and the MCS performance estimate. The performance estimate (MCS standard MIS 3002) records the system size (kWp) and each array's inclination (tilt), orientation and shading factor. The installer named on the Sunsynk plant record is shown on the Data & trust page.
2. **Read the figures:**
   - system size, kWp → `PV_KWP`;
   - inclination, degrees from flat → `PV_TILT`;
   - orientation → `PV_ORIENTATION`, as a compass bearing (east 90, south 180, west 270). If it is given as degrees away from south, use 180 minus it for east of south and 180 plus it for west of south;
   - the battery's usable capacity, kWh, from its datasheet or the handover pack → `BATTERY_KWH`.

3. **Put them in `.env` and restart** (`npm start`). On the next sync:
   - the configured values replace the estimates (`.env` always takes priority);
   - Quartz forecasts for those panels are compared with actual output for 2 days before they are relied on.
4. **If the panels face more than one way,** note each array's figures. The prototype models one direction. That is right for this home: both strings produce equal power.

Other ways to confirm the values, in order of reliability:

1. **The installation paperwork,** as above.
2. **Measure on site:** a compass app for direction and an inclinometer app on the panel frame for tilt.
3. **Re-run the estimate** with more days, especially clear ones.

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8. Proving the data: the Data & trust page

The pages show conclusions (costs, savings, charts). The Data & trust page shows why they can be relied on, from the facts themselves. Every view below is computed on request from stored facts (`/api/sites/:id/trust`, `/revisions`, `/asof`); none of it is typed in. Every other page carries a one-line note of what its figures rest on for the selected period (for example "315 half-hours: 77% final, 73 provisional · 70 of 72 cross-checks agree"), linking here.

8.1 Headline

For the selected period:

- **Final share:** half-hours final, provisional and missing.
- **Cross-checks:** passed, warnings, failed, not yet judged.
- **Traced:** provider facts in the period that link to the raw response they came from (100% on 16 Oct: 14,880 facts from 127 responses).
- **Revised:** half-hours changed after they were first shown as final (the `revised` quality bit), days with a recorded revision, and corrections by EnergieBee.

8.2 How the timeline comes together (source coverage)

One row per source, one cell per half-hour, plus the build's result (final, provisional, missing) as the last row. A cell's state comes from the facts' own timestamps: the first version's `recorded_at` against the end of the half-hour.

Source	Expected per half-hour	Counts as late after
Inverter 5-minute power (Sunsynk)	6 samples (fewer than 5 = partial)	1 hour
Smart meter import, export, gas (Octopus)	1 reading	2 days (Octopus delivers about a day late)
Gas calorific value (National Gas)	1 value per gas day	2 days
Grid carbon (NESO), sunlight (Open-Meteo)	1 value	1 day
Import and export price	a price in force	— (published ahead)
Solar forecast made in advance	a forecast issued before the half-hour	— (kept from 16 Oct)

"Loaded with the history" marks data fetched in the first 15 minutes after a source was first connected, for half-hours that had already ended: the initial 14-day load, not lateness. "Waiting" means not received and not yet due.

8.3 Independent measurements agree (cross-check grid)

Days across, checks down, each cell with its status and measured value (hover for the exact numbers). The checks are those in section 1.3: the six inverter counters against the 5-minute samples, smart meter against inverter clamp for import and export, the energy balance against this home's typical day, coverage and pricing.

8.4 Every kWh accounted for

For each day, where the energy came from (solar, grid, battery) and where it went (house, exports, into the battery, the inverter's own use), drawn to the same scale. In always equals out, because each source's energy is split into flows and what is left after the house, exports and charging is the inverter's use. Next to it: how many inverter counters agreed and the smart-meter against clamp totals.

Two figures for the inverter's use. `site_hh.loss` is the energy balance minus the inverter's load reading; it goes negative in half-hours where the load reading exceeds the balance (on 14 Oct the sum was 2.9 kWh). The `*_to_loss` flows are what each source supplied beyond the house, exports and charging (4.0 kWh that day); they are what every page prices and what the balance check now uses.

8.5 How each day changed (revision log)

`site_day` keeps only the latest version of a day. Each time a completed day is rebuilt with different figures, state or check results, a row is added to `site_day_revision`: when, why (the reason that queued the rebuild, such as "facts changed" or "full rebuild requested"), what changed (from → to), the build method, and the inputs (battery size and its source, half-hours from the meter and from the clamp). An unchanged rebuild adds nothing. Every row has a kind:

- **Recorded:** written by a live rebuild. The live log started at 13:53 BST on 16 Oct 2026.
- **Reconstructed:** added afterwards for the time before the live log, by `npm run history:reconstruct` (`src/core/history.js`).
  - **How:** each completed day is rebuilt "as known at" (section 8.6) the end of every earlier sync. A version is stored only when it differs from the previous one.
  - **Inputs:** each row lists any input read as first recorded (section 8.6).
  - **Link to the live log:** each day's first recorded row then says what changed since the last reconstructed version.
  - **Run:** it runs once (`--force` redoes it) and writes only to `site_day_revision`.
- **What a reconstructed version is:** the figures today's build method gives from the facts known at that moment, not a copy of what the pages showed then. The method itself changed during 16 Oct.
- **Where it starts:** 07:49 BST on 16 Oct, the first sync whose tariff agreements were recorded. Before that the tariff came from `.env` and was not kept, so a bill cannot be reproduced.
- **Example, 14 Oct (10 Acresbrook):**

Time (BST)	Kind	Version
07:49	reconstructed	48 half-hours final (meter readings in), gas unknown, bill not priced
10:12	reconstructed	gas 37.3 kWh: the m³ readings and calorific value were stored

Time (BST)	Kind	Version
12:01	reconstructed	bill £4.46: Octopus's time-stamped IOG prices arrived, so the off-peak window could be learned
13:53	recorded	no change

For 10 Acresbrook: 64 syncs replayed in 9.4 s, giving 42 versions across 14 days.

8.6 A day as it was known at any moment ("as known at")

Facts are never overwritten, so the build can run as it would have at an earlier moment T. Every fact, price and system detail is read in the version in force at T (`recorded_at ≤ T` and not yet superseded), forecasts count only if issued by T, and "now" is T. The rebuild runs from the first day (so the battery stock is also as known then), writes nothing, and is set beside the day as it is now.

- **Checked:** "as known now" reproduces the stored rows exactly (698 half-hours, 0 differences; a test).
- **Example:** 15 Oct as known at 08:00 BST on 16 Oct had 46 provisional half-hours from the clamp and no bill, because the Intelligent Octopus Go off-peak window could not yet be learned. Now it has a bill of £4.70.
- **What is read as of T:**
  - facts and prices;
  - tariff agreements: versioned, a changed account history supersedes the old rows;
  - meters: a device has a first-known time and is retired, not deleted;
  - device details (gas unit, gas zone): versioned in `device_attr_version`;
  - system details.
- **Limits:** where no version existed yet at T, the earliest recorded one is used and listed with the result as "read as first recorded". For 10 Acresbrook:
  - device details are versioned from 14:28 BST and system details from 13:53 BST on 16 Oct;
  - when devices and agreements were first known comes from the sync log: the first sync whose Octopus step succeeded, 07:49 BST;
  - the `.env` tariff used before that was not kept;
  - before the first sync nothing was known, so such a moment shows nothing.

8.7 System details, meters and tariffs, versioned

Panel size, tilt, direction and battery size are kept in `site_attribute`, one current value per source: installer (`.env`), measured (from the home's data), provider (Sunsynk) or estimated. A new value supersedes the same source's previous one; nothing is deleted. The value used is the highest-ranked source in force: installer, then measured, then provider, then estimated. The site row caches it. "Your system's details" shows each value with its source, and every recorded version.

Meters and tariffs follow the same rule:

- **Device details** (gas unit, gas zone, EV status) get a new `device_attr_version` row when they change. `device.attrs` holds the current one.

- **Devices** that stop being used are retired (`retired_at`). For example, the stand-in `.env` tariff meters are retired once the Octopus meters are read.
- **Agreements:** a changed account history supersedes the old tariff agreements (kept) and adds the new ones.
- **Timestamps:** a version's time always moves forward, even for two changes in the same millisecond.
- **Upgrade:** a database from before this upgrades itself on open. The agreement table is rebuilt once, and the first-known times are filled from the sync log.

8.8 Corrections by EnergieBee

When the core finds it stored something a provider did not send, it withdraws it (supersedes, never deletes) and logs it in `fact_correction`.

- **16 Oct 2026:** 626 `gas_kwh` facts labelled as Octopus readings were withdrawn. Earlier code had converted the m³ readings with a fixed calorific value as it stored them (0.118 m³ × 11.22 = 1.324 kWh). Octopus sent m³; those readings are kept as `gas_m3` and converted in the build with the daily value. The build had already ignored them, so no figure changed (a test checks this).

8.9 The path of one half-hour

For any half-hour, seven steps from left to right:

1. the provider responses (each opens as received);
2. the facts, with older versions;
3. align (fit factors to the daily counters);
4. reconcile (meter or clamp, with the clamp's reading kept when the meter replaced it);
5. allocate (the flows);
6. price (tariff, register, rate with VAT, bill);
7. check (state and the day's checks).

The full numbers, facts and prices in force follow below.

Sources

- Octopus Energy: How does Intelligent Octopus Go work; Intelligent Octopus Go charge limit (7 May 2026)
- National Gas: Gas Data Portal; data item "Calorific Value, LDZ(NW)" (PUBOB4512) via `https://data.nationalgas.com/api/find-gas-data-download`
- Gas day 05:00–05:00: UNC Modification 0461; SI 2015/953
- Ofgem: Review of typical domestic consumption values: decision (May 2026)